

Annex G (informative)

Examples

The following examples are provided for the purpose of software validation. The number of digits is not indicative of the uncertainty in the calculation, but is only given for testing purposes.

Table G.1 — Gas analysis by mole fraction

	Component	Gas 1	Gas 2	Gas 3	Gas 4	Gas 5	Gas 6
1	Methane	0,796	0,650	0,720	0,550		
2	Nitrogen	0,100	0,065	0,010		0,25	
3	Carbon dioxide	0,010	0,190	0,010		0,15	
4	Ethane	0,057	0,010	0,150			
5	Propane	0,020	0,010	0,010	0,100		0,10
6	n-Butane	0,005	0,010	0,010			0,10
7	iso-Butane	0,005	0,010	0,010			0,10
8	n-Pentane	0,002	0,010	0,010			0,10
9	iso-Pentane	0,002	0,010	0,010			0,10
10	n-Hexane	0,001	0,010	0,020			0,10
11	n-Heptane	0,001	0,020	0,010			0,10
12	n-Octane	0,001		0,010			0,10
13	n-Nonane						0,10
14	n-Decane						0,05
15	Hydrogen				0,200	0,10	
16	Oxygen			0,010	0,049 99	0,10	
17	Carbon monoxide			0,010	0,100	0,10	
18	Water				0,000 01		0,05
19	Hydrogen sulfide					0,10	
20	Helium		0,005			0,10	
21	Argon					0,10	
	Total	1,000	1,000	1,000	1,000	1,000	1,000

Table G.2 — Calculations

Gas 1		$M = 19,778 \text{ kg/kmol}$								
T	p	p	Z	U	H	S	C_v	C_p	w	μ
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
180,0	10,0	389,18	0,33956	-599,26	-573,56	-3,8855	1,6233	3,3622	796,80	0,0080269
220,0	10,0	267,65	0,40397	-450,10	-412,74	-3,0842	1,6813	5,0560	427,36	1,8447
200,0	20,0	380,27	0,62552	-560,41	-507,82	-3,6760	1,6149	3,1519	826,44	-0,052376
250,0	20,0	283,04	0,67232	-411,32	-340,66	-2,9315	1,6215	3,4875	568,65	0,77140

Table G.2 (continued)

305,0	3,0	24,835	0,94211	-135,65	-14,850	-1,1125	1,5553	2,1284	394,93	4,5112
350,0	10,0	74,667	0,91021	-100,62	33,305	-1,4321	1,7043	2,5065	430,98	2,6236
Gas 2		M = 26,843 kg/kmol								
<i>T</i>	<i>p</i>	<i>ρ</i>	<i>Z</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>C_v</i>	<i>C_p</i>	<i>w</i>	<i>μ</i>
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
180,0	13,0	548,61	0,42502	-539,66	-515,96	-3,0347	1,3262	2,4027	945,53	-0,28327
220,0	11,0	452,06	0,35708	-436,41	-412,08	-2,4955	1,3851	2,8949	615,88	0,26965
250,0	20,0	425,35	0,60721	-381,58	-334,56	-2,2525	1,4023	2,7341	615,09	0,26645
350,0	20,0	226,57	0,81426	-141,22	-52,944	-1,3037	1,5242	2,6216	414,60	1,6170
355,0	3,0	28,743	0,94921	-32,462	71,912	-0,43567	1,4383	1,8513	357,41	4,0492
400,0	10,0	88,041	0,91676	4,0059	117,59	-0,66139	1,5813	2,1613	383,33	2,4644
Gas 3		M = 24,295 kg/kmol								
<i>T</i>	<i>p</i>	<i>ρ</i>	<i>Z</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>C_v</i>	<i>C_p</i>	<i>w</i>	<i>μ</i>
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
150,0	10,0	504,68	0,38600	-711,53	-691,71	-3,9440	1,6072	2,6157	1267,9	-0,46108
200,0	10,0	431,60	0,33852	-577,62	-554,45	-3,1562	1,6072	2,9153	889,61	-0,17932
250,0	20,0	376,46	0,62095	-456,79	-403,67	-2,5938	1,6569	3,0291	731,32	0,072674
300,0	20,0	289,77	0,67228	-312,74	-243,72	-2,0114	1,7557	3,3290	523,24	0,83762
380,0	5,0	41,741	0,92111	4,6359	124,42	-0,54631	1,8869	2,4301	378,29	3,5788
400,0	10,0	82,020	0,89066	13,893	135,81	-0,73163	1,9839	2,6995	392,34	2,6674
Gas 4		M = 18,037 kg/kmol								
<i>T</i>	<i>p</i>	<i>ρ</i>	<i>Z</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>C_v</i>	<i>C_p</i>	<i>w</i>	<i>μ</i>
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
180,0	25,0	378,94	0,79511	-587,59	-521,61	-3,9519	1,6680	3,1029	849,33	-0,15641
220,0	25,0	307,84	0,80080	-474,56	-393,34	-3,3091	1,6573	3,2832	660,81	0,27097
240,0	1,7	16,192	0,94898	-240,29	-135,30	-1,2079	1,5070	2,0930	371,93	5,5555
260,0	25,0	242,33	0,86077	-365,93	-262,77	-2,7635	1,6780	3,1985	564,70	0,69308
300,0	10,0	80,581	0,89738	-193,05	-68,953	-1,7099	1,6729	2,5985	425,19	2,6008
400,0	10,0	54,906	0,98774	1,9608	184,09	-0,98172	1,8988	2,5469	495,53	1,3590
Gas 5		M = 27,610 kg/kmol								
<i>T</i>	<i>p</i>	<i>ρ</i>	<i>Z</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>C_v</i>	<i>C_p</i>	<i>w</i>	<i>μ</i>
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
150,0	30,0	704,83	0,94229	-353,11	-310,55	-2,4550	0,89982	1,7205	713,43	-0,26388
200,0	25,0	502,27	0,82643	-269,63	-219,85	-1,8878	0,84707	1,8818	485,49	0,41958
250,0	20,0	299,16	0,88802	-188,56	-121,70	-1,3922	0,81223	1,6418	391,32	1,3328

Table G.2 (continued)

300,0	15,0	172,80	0,96087	-123,76	-36,952	-1,0017	0,79210	1,3398	387,22	1,5610
350,0	10,0	95,443	0,99409	-69,180	35,595	-0,65789	0,78914	1,1995	401,08	1,4409
400,0	5,0	41,380	1,0031	-19,179	101,65	-0,27251	0,79493	1,1359	416,75	1,2625
Gas 6		$M = 81,365 \text{ kg/kmol}$								
<i>T</i>	<i>p</i>	<i>p</i>	<i>Z</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>C_v</i>	<i>C_p</i>	<i>w</i>	<i>p</i>
K	MPa	kg/m ³		kJ/kg	kJ/kg	kJ/(kg·K)	kJ/(kg·K)	kJ/(kg·K)	m/s	K/MPa
180,0	10,0	758,08	0,71716	-612,96	-599,77	-1,9233	1,4744	1,9300	1603,4	-0,54380
250,0	10,0	699,02	0,55998	-474,26	-459,95	-1,2685	1,6214	2,0839	1283,4	-0,48029
300,0	25,0	671,21	1,2150	-374,55	-337,30	-0,90152	1,7870	2,2353	1184,5	-0,43573
400,0	5,0	549,38	0,22266	-114,62	-105,52	-0,14589	2,1625	2,7652	645,44	-0,091974
450,0	0,5	11,811	0,92057	247,23	289,56	0,86891	2,2505	2,4030	203,37	11,662
500,0	2,0	52,175	0,75024	337,90	376,23	0,93137	2,5111	2,8942	176,26	11,031

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